

Task 12: Support Vector Machines (SVM)

Objective

Build and compare SVM models using:

Linear Kernel

Polynomial Kernel

RBF Kernel

Dataset: Social Network Ads Dataset

Final Outcome: Understand:

Kernel Trick

Hard vs Soft Margin

Role of C and Gamma

Interview questions

1. What is a Hyperplane?

A hyperplane is the decision boundary that separates different classes in SVM.

For 2D data:

Hyperplane = Line

For 3D data:

Hyperplane = Plane

For higher dimensions:

Generalized boundary

Example:



2. Explain Kernel Trick

The Kernel Trick helps SVM classify non-linear data by transforming it into higher dimensions where it becomes linearly separable.

Common kernels:

Linear Kernel

Polynomial Kernel

RBF (Radial Basis Function)

Example of polynomial mapping:



3. Hard Margin vs Soft Margin

Hard Margin

No misclassification allowed

Works only for perfectly separable data

Soft Margin

Allows small classification errors

More practical for real-world datasets

4. Role of C and Gamma

C Parameter

Controls penalty for misclassification.

High C:

Less misclassification

Risk of overfitting

Low C:

More flexible boundary

Better generalization

Gamma Parameter (RBF Kernel)

Controls influence of each training point.

High Gamma:

Complex boundary

Overfitting risk

Low Gamma:

Smooth boundary

Underfitting risk

Expected Output

You will get:

Accuracy scores

Confusion matrices

Classification reports

Hyperplane visualizations

Kernel comparison graph

Conclusion

In this task, you learned:

SVM classification

Feature scaling importance

Kernel Trick

Linear vs Non-linear classification

Hyperplane visualization

Effect of C and Gamma

Kernel performance comparison